Piri Reis University

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GRADUATION THESIS – YBS421

SUPPLY CHAIN VISIBILITY AND RISK ASSESSMENT PLATFORM

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DECLARATION

I hereby declare that this thesis represents my own work which has been done after registration for the bachelor's degree at Piri Reis University and has not been previously included in a thesis or dissertation submitted to this or any other institution for a degree, diploma or other qualifications.

Signature:

Date:

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ABSTRACT

Supply chains include many connected activities such as order management, shipment planning, inventory control, delivery tracking, regional performance monitoring, and demand planning. When these activities are analyzed separately, managers may have difficulty seeing operational problems and risks on time. This graduation thesis develops a Supply Chain Visibility and Risk Assessment Platform to support supply chain decision-making through data visualization, risk analysis, and dataset standardization.

The platform was developed with Python and Streamlit. It uses the public DataCo Smart Supply Chain dataset as the default demonstration dataset. In addition, the system allows users to upload a new supply chain dataset in CSV or Excel format. The uploaded dataset can be mapped to a standard supply chain schema and then used dynamically across the dashboard pages. The platform includes modules for overview analysis, regional performance, inventory control, shipment performance, supply chain risk, and demand forecasting.

The main contribution of the project is that it is not limited to a single static dataset. Instead, it provides a reusable dashboard structure where different supply chain datasets can be standardized and analyzed. The results show that a dataset-adaptive dashboard can improve visibility and help users identify delivery risks, inventory warnings, regional issues, and demand patterns from a single interface.

Key Words: Supply Chain Visibility, Risk Assessment, Data Standardization, Streamlit Dashboard, Business Intelligence, Demand Forecasting

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1 INTRODUCTION

Supply chain management includes many connected activities such as order processing, shipment planning, inventory control, delivery tracking, regional performance monitoring, and demand planning. In many organizations, these activities are recorded in different files, systems, or reports. When supply chain data is not analyzed in an integrated way, managers may have difficulty understanding current performance and detecting operational risks on time.

Visibility is one of the most important needs in supply chain operations. Managers need to see how orders, shipments, products, regions, inventory indicators, and demand patterns are performing. Without a clear visibility system, late deliveries, stock risks, regional problems, and demand changes may not be noticed early enough. For this reason, dashboard-based decision-support systems are useful for transforming raw supply chain data into understandable information.

This graduation thesis develops a Supply Chain Visibility and Risk Assessment Platform. The aim of the project is to create an interactive system that helps users monitor supply chain performance and operational risk from a single interface. The platform provides dashboard modules for executive overview, regional performance, inventory control, shipment performance, supply chain risk analysis, and demand forecasting.

The project uses the public DataCo Smart Supply Chain dataset as the default demonstration dataset. However, the system is not limited to this dataset. Users can also upload their own supply chain datasets in CSV or Excel format. After uploading a dataset, users map its columns to a standard supply chain schema. The system then standardizes the uploaded data and dynamically updates the dashboard pages according to the active dataset.

The main purpose of this project is not only to visualize one historical dataset, but also to design a reusable dashboard structure for different supply chain datasets. By combining data upload, column mapping, data standardization, KPI calculation, risk scoring, and forecasting, the platform supports a more flexible approach to supply chain visibility and risk assessment.

2 LITERATURE REVIEW

This section reviews the main concepts related to the project. The literature review focuses on supply chain visibility, business intelligence dashboards, supply chain risk management, digital transformation in logistics, demand forecasting, and data standardization. These topics are directly related to the development of a dataset-adaptive supply chain visibility and risk assessment platform.

2.1 Supply Chain Visibility

Supply chain visibility refers to the ability to monitor and understand the movement of products, orders, shipments, inventory, and related information across the supply chain. In modern supply chains, operations are often spread across different regions, suppliers, transport modes, and information systems. Because of this complexity, managers need integrated visibility tools to understand what is happening in the supply chain and to detect problems early. (Oracle, 2023)

Visibility is important because supply chain decisions depend on timely and accurate information. If delivery delays, inventory shortages, regional problems, or demand changes are not visible, managers may react too late. A visibility system can help users monitor key indicators such as order volume, delivery status, shipment delays, inventory condition, sales, profit, and demand trends from a single interface. (Oracle, 2023; Kalaiarasan, Olhager, Agrawal, & Wiktorsson, 2022)

In this project, supply chain visibility is addressed through a multi-page dashboard structure. The platform brings different operational views together, including overview indicators, regional performance, inventory control, shipment performance, risk analysis, and demand forecasting. This supports the idea that supply chain visibility should not focus on only one metric, but should combine different operational indicators.

2.2 Business Intelligence Dashboards and Decision Support

Business intelligence dashboards are used to transform raw data into visual and understandable information. Instead of analyzing large tables manually, users can monitor key performance indicators, charts, and summary tables through an interactive interface. Dashboards are especially useful in operational areas where managers need to make quick and data-based decisions. (Grabińska, 2019)

In supply chain management, dashboards can support decision-making by showing delivery performance, regional differences, inventory problems, shipment delays, and risk indicators. A well-designed dashboard does not only display data; it also helps users compare performance, identify patterns, and focus on problem areas. For this reason, dashboards can act as decision-support tools for supply chain managers, logistics analysts, inventory planners, and business analysts. (Grabińska, 2019)

In this context, business intelligence dashboards support managerial decision-making by transforming raw operational data into structured indicators, visual summaries, and performance monitoring tools. (Grabińska, 2019)

This project uses a dashboard-based approach because the goal is to make supply chain data more usable for decision-making. The system includes interactive filters, KPI cards, charts, tables, risk scores, and forecast outputs. These features help users move from raw data to operational insight.

Recent studies also emphasize that supply chain visibility improves coordination, responsiveness, and decision-making by making operational information more accessible across the supply chain network. (Kalaiarasan, Olhager, Agrawal, & Wiktorsson, 2022)

2.3 Supply Chain Risk Management

Supply chain risk management focuses on identifying, assessing, and reducing risks that can affect supply chain performance. These risks may include late deliveries, transportation delays, demand uncertainty, inventory shortages, regional disruptions, supplier problems, and financial losses. Since supply chains include many connected activities, a problem in one part of the system can affect other parts of the operation. (Das, Kumar, & Tiwari, 2025)

Risk assessment is more useful when different risk indicators are evaluated together. For example, late delivery rate alone may not fully explain supply chain risk. A better risk view may also include delay days, inventory condition, demand volatility, and regional performance. Therefore, dashboard-based risk analysis can help managers understand not only where risks exist, but also which indicators contribute to those risks. (Das, Kumar, & Tiwari, 2025)

In this project, supply chain risk is calculated using rule-based and interpretable indicators. The system analyzes delivery risk, delay risk, inventory risk, demand volatility risk, and region-level risk. The purpose is not to create a complex black-box prediction model, but to provide a clear and understandable risk assessment structure for decision support.

For this reason, interpretable risk indicators are useful in operational dashboards because they allow managers to identify problematic areas such as late deliveries, inventory shortages, and regional performance issues without relying on black-box models. (IBM, 2024)

2.4 Digital Transformation and Data-Driven Logistics

Digital transformation in logistics and supply chain management is closely related to the use of data, automation, analytics, and connected systems. Digital tools can improve transparency by collecting and analyzing operational data from different stages of the supply chain. This can support better planning, faster decision-making, and improved coordination between supply chain activities. (Raza, Svanberg, & Wiegmans, 2023)

However, digital transformation also requires data integration. Many organizations keep supply chain information in different systems, files, or formats. If these datasets cannot be combined or standardized, it becomes difficult to create useful dashboards and risk analysis tools. For this reason, data standardization is an important part of supply chain analytics. (IBM, 2024)

This project addresses this issue by adding a dataset upload and standardization module. Users can upload a new CSV or Excel supply chain dataset and map its columns to a standard schema. After this mapping process, the system converts the uploaded data into dashboard-ready analytical tables. This makes the platform more flexible than a dashboard that works with only one fixed dataset.

2.5 Demand Forecasting in Supply Chain Planning

Demand forecasting is an important part of supply chain planning because it helps organizations estimate future demand and prepare operational plans. Forecasting can support decisions related to inventory levels, replenishment planning, transportation capacity, and product availability. Even simple forecasting methods can be useful for understanding general demand trends. (Hyndman & Athanasopoulos, 2021)

In supply chain dashboards, demand forecasting is often used together with historical demand analysis. Users can compare past demand patterns with forecasted values and evaluate whether demand is increasing, decreasing, or unstable. Forecast accuracy metrics can also help users understand how well a forecasting method performs. (Hyndman & Athanasopoulos, 2021)

In this project, the demand forecasting module uses historical monthly demand data. The platform includes simple forecasting methods such as moving average and linear regression. These methods are suitable for a dashboard prototype because they are understandable, easy to implement, and useful for demonstrating how demand data can be turned into future demand estimates.

Moving average and regression-based models are commonly used as baseline forecasting approaches because they are simple, interpretable, and suitable for demonstrating demand trends in decision-support systems. (Hyndman & Athanasopoulos, 2021)

2.6 Data Standardization and Uploaded Dataset Support

A major challenge in data-driven dashboard systems is that datasets may have different column names, formats, and levels of completeness. For example, one dataset may use “order date,” another may use “PO created date,” and another may use a different name for the same concept. If the system cannot handle these differences, it can only work with one fixed dataset. (IBM, 2024)

To make a dashboard reusable, uploaded datasets should be converted into a common structure. This requires a standard schema, column mapping, data validation, and derived indicators. In this project, the standard schema includes fields such as order ID, order date, product name, category, quantity, shipping date, scheduled delivery date, delivery status, shipping mode, market, region, country, sales, and profit.

The dataset upload feature improves the originality and usefulness of the project. The default DataCo dataset is used as a demonstration dataset, but the system can also process new supply chain datasets. After the user maps the uploaded dataset to the standard schema, the dashboard pages are generated dynamically based on the active dataset. This makes the platform more reusable for different supply chain data sources.

The DataCo supply chain dataset is suitable for this prototype because it contains order, customer, product, shipment, sales, and delivery-related fields that can be transformed into dashboard-ready analytical tables. (Kaggle, n.d.)

2.7 Summary of Literature Review

The reviewed concepts show that supply chain visibility, dashboard-based decision support, risk assessment, digital transformation, forecasting, and data standardization are strongly connected. Supply chain visibility helps managers monitor operational activities. Business intelligence dashboards make complex data easier to understand. Risk management supports the identification of delivery, inventory, demand, and regional risks. Forecasting helps users observe future demand trends. Data standardization allows different datasets to be used in a common dashboard structure.

Based on these ideas, this project develops a dataset-adaptive supply chain visibility and risk assessment platform. The platform combines dashboard visualization, uploaded dataset support, data standardization, risk scoring, and demand forecasting in a single Streamlit-based system.

3 ORIGINALITY OF THE STUDY

The originality of this study comes from developing a supply chain dashboard platform that is not limited to one fixed dataset. Many dashboard projects are designed only for a single prepared dataset. In this project, the DataCo Smart Supply Chain dataset is used as the default demonstration dataset, but the platform also allows users to upload a new supply chain dataset in CSV or Excel format.

The system includes a Data Source Manager that supports column mapping and dataset standardization. This means that different datasets with different column names can be mapped to a common supply chain schema. After the mapping process, the system converts the uploaded dataset into standardized analytical tables and uses these tables across all dashboard pages.

Another original aspect of the project is the integration of multiple supply chain analysis areas in one platform. The system does not only show order or sales information. It combines executive KPIs, regional performance analysis, inventory control indicators, shipment performance, supply chain risk scoring, and demand forecasting. This makes the project closer to a decision-support platform rather than a simple visualization page.

The risk analysis structure is also designed to be understandable for users. Instead of using a black-box model, the project uses rule-based and interpretable risk indicators such as late delivery rate, delay days, inventory status, demand volatility, and regional performance. This helps users understand why a specific market, region, shipment mode, or product category may be risky.

The project is original within its academic scope because it combines data upload, schema mapping, data standardization, dynamic dashboard generation, risk analysis, and forecasting in a single Streamlit-based system. Therefore, the main contribution of the project is the development of a reusable supply chain visibility and risk assessment platform that can work with both a default dataset and user-uploaded datasets.

4 PROBLEM DEFINITION

Supply chain operations produce large amounts of data related to orders, products, shipments, delivery performance, inventory, regions, sales, and demand. However, this data is often stored in different files, systems, or formats. When the data is not standardized and analyzed together, it becomes difficult for managers to understand the overall performance of the supply chain.

One of the main problems is the lack of integrated visibility. A manager may need to check different reports to understand late deliveries, shipment delays, inventory warnings, regional performance, and demand changes. This slows down decision-making and increases the risk of missing important operational problems.

Another problem is that many dashboard systems are designed for a single prepared dataset. If a new dataset has different column names or a different structure, the dashboard may not work without manual code changes. This limits the reusability of the system. For example, one dataset may use “order date,” while another dataset may use “PO created date” for the same concept. Without a standard mapping and transformation process, these datasets cannot be analyzed through the same dashboard structure.

There is also a risk assessment problem. Supply chain risk should not be evaluated by looking at only one indicator. Late delivery rate, average delay days, inventory status, demand volatility, regional performance, and financial indicators can all affect operational risk. If these indicators are not combined in a clear way, users may not understand which areas of the supply chain require attention.

Therefore, the main problem addressed in this project is the need for a reusable and interactive decision-support platform that can transform different supply chain datasets into a standard analytical format and generate visibility dashboards and risk analysis from the active dataset.

5 SOLUTION APPROACH

The solution approach of this project is based on developing a dataset-adaptive supply chain dashboard platform. The system is designed to work with a default demonstration dataset and also with user-uploaded supply chain datasets. The main idea is to transform different supply chain datasets into a common analytical structure and then generate dashboard modules from the active dataset.

5.1 Overall Approach

The project follows a data-driven decision-support approach. First, supply chain data is collected or uploaded into the system. Then, the data is cleaned, standardized, and transformed into dashboard-ready analytical tables. After this transformation, the dashboard modules calculate KPIs, generate visualizations, and support risk assessment.

The platform uses the DataCo Smart Supply Chain dataset as the default dataset. This dataset is processed into different analytical tables such as orders, products, shipments, inventory, demand, and risk scores. These tables are used by the default dashboard when no uploaded dataset is active.

However, the project is not limited to the default dataset. Users can upload a new company supply chain dataset in CSV or Excel format. The uploaded dataset is mapped to a standard schema through the Data Source Manager. After the mapping process, the system creates the same type of analytical tables and updates the dashboard pages according to the uploaded dataset.

5.2 Dataset Upload and Standardization

The dataset upload feature is one of the main parts of the solution. Different supply chain datasets may use different column names for similar concepts. For example, one dataset may use “order date,” another may use “PO created date,” and another may use “created at.” To solve this problem, the platform includes a column mapping process.

In the Data Source Manager, users map the columns of the uploaded dataset to the standard supply chain schema. The required fields include order ID, order date, product name, category name, and order quantity. Optional fields include shipping date, scheduled delivery date, delivery status, actual shipping days, scheduled shipping days, shipping mode, market, region, country, sales, and profit.

After the mapping is completed, the system standardizes the uploaded dataset. It converts date fields, numeric fields, and text fields into a common format. It also derives additional indicators where possible. For example, if actual shipping days and scheduled shipping days are not available, the system can calculate delay days by comparing delivered date and scheduled delivery date.

5.3 Dynamic Dashboard Generation

After the dataset is standardized, the dashboard pages are generated from the active dataset. If no uploaded dataset is active, the dashboard uses the default DataCo dataset. If the user uploads and activates a new dataset, all dashboard pages use the uploaded dataset during the active Streamlit session.

The dashboard includes six main modules: Overview, Regional Performance, Inventory Control, Shipment Performance, Supply Chain Risk, and Demand Forecast. Each module focuses on a different part of supply chain visibility. The Overview page shows general KPIs and summary charts. The Regional Performance page compares markets and regions. The Inventory Control page shows generated inventory indicators. The Shipment Performance page analyzes delivery and delay performance. The Supply Chain Risk page calculates risk indicators. The Demand Forecast page analyzes historical demand and creates simple forecasts.

This structure allows the system to work as a flexible platform instead of a static dashboard. The same dashboard pages can be used with different supply chain datasets as long as the required fields are mapped correctly.

5.4 Risk Assessment Approach

The risk assessment approach is rule-based and interpretable. The purpose of the risk model is not to create a complex black-box prediction system. Instead, the aim is to provide clear risk indicators that users can understand and discuss.

The platform uses different rule-based risk scores for different analysis levels, including overall supply chain risk, regional risk, and category-level risk. The overall risk score combines delivery risk, inventory risk, delay risk, and demand volatility risk, while regional and category-level scores use related operational indicators such as late delivery rate, delay days, profit risk, and inventory condition.

This approach is suitable for a graduation thesis prototype because it provides a transparent decision-support structure. Users can understand which factors affect the risk score and can use the dashboard to explore risky regions, shipment modes, categories, and inventory items.

5.5 Forecasting Approach

The platform also includes a demand forecasting module. This module uses historical monthly demand values and applies simple forecasting methods such as moving average and linear regression. The purpose is to show how historical demand data can be used to estimate future demand patterns.

The forecasting module also includes basic performance evaluation metrics such as MAE and MAPE. These metrics help users compare forecast accuracy and understand the reliability of the forecast results. Since the project focuses mainly on supply chain visibility and risk assessment, the forecasting module is designed as a planning-support feature rather than an advanced forecasting system.

6 REQUIRED TOOLS

This project was developed by using Python-based data analysis and dashboard development tools. The selected tools were chosen because they are open-source, widely used, and suitable for building a supply chain analytics prototype.

6.1 Programming and Development Environment

Python was used as the main programming language of the project. It was used for data cleaning, data transformation, KPI calculation, risk score generation, demand forecasting, and dashboard development. Visual Studio Code was used as the development environment for writing and editing the project files.

The project was developed as a local Streamlit application. A Windows batch file was also created to make the project easier to run. This file automatically creates the virtual environment if needed, installs the required packages, processes the default dataset, and starts the dashboard.

6.2 Data Processing and Analysis Tools

pandas was used for reading CSV and Excel files, cleaning the dataset, grouping records, calculating KPIs, and creating analytical tables. NumPy was used for numerical calculations and handling missing numerical values. (pandas development team, n.d.)

SQLite was used as a lightweight database solution for storing the processed default dataset. The processed data files were also saved as CSV files so that the dashboard could load them easily.

6.3 Dashboard and Visualization Tools

Streamlit was used to develop the interactive dashboard interface. It allowed the project to include different pages, filters, file upload components, KPI cards, charts, and tables.

Plotly was used for creating interactive charts such as bar charts, line charts, heatmaps, and risk visualizations. These visualizations help users understand supply chain performance and risk indicators more easily.

6.4 Forecasting and Version Control Tools

scikit-learn was used for the linear regression model in the demand forecasting module. The project also uses a simple moving average method to compare forecasting results.

Git and GitHub were used for version control, project backup, and documentation. GitHub also makes it easier to share the project files, screenshots, diagrams, and source code.

Table 1: Required Tools and Technologies Used in the Project

Tool / Library	Type	Purpose	Used In
Python	Programming Language	Data processing, risk scoring, forecasting, dashboard logic	Whole project
Streamlit	Dashboard Framework	Interactive dashboard pages, filters, file upload, KPI cards	Dashboard interface
pandas	Data Analysis Library	Reading datasets, cleaning data, grouping, KPI calculation	Data processing
NumPy	Numerical Library	Numerical operations and missing value handling	Data processing
Plotly	Visualization Library	Interactive charts and dashboard visuals	Dashboard pages
scikit-learn	Machine Learning Library	Linear regression forecasting model	Demand Forecast module
SQLite	Database	Lightweight storage for processed default data	Processed data storage
Git / GitHub	Version Control	Project backup, version tracking, sharing project files	Documentation and development

7 SYSTEM / MODEL & SOLUTION DEVELOPMENT

This section explains the technical development of the Supply Chain Visibility and Risk Assessment Platform. The system was designed as a dataset-adaptive dashboard platform. It can work with the default DataCo dataset, and it can also process user-uploaded supply chain datasets through column mapping and data standardization.

The system development process includes data source selection, dataset upload, column mapping, data standardization, analytical table generation, dashboard module development, risk score calculation, and demand forecasting.

7.1 System Architecture

The platform consists of several connected layers. The first layer is the data source layer. This layer includes the default DataCo dataset and user-uploaded CSV or Excel datasets. The second layer is the Data Source Manager, where the user can upload a new dataset and select the active data source. The third layer is the column mapping and data adapter layer. This layer converts different dataset structures into a common supply chain schema.

After the data is standardized, the system creates dashboard-ready analytical tables such as orders, products, shipments, inventory, demand, and risk scores. These tables are then used by the Streamlit dashboard modules. The main dashboard modules are Overview, Regional Performance, Inventory Control, Shipment Performance, Supply Chain Risk, and Demand Forecast.

The system architecture is shown in Figure 1.

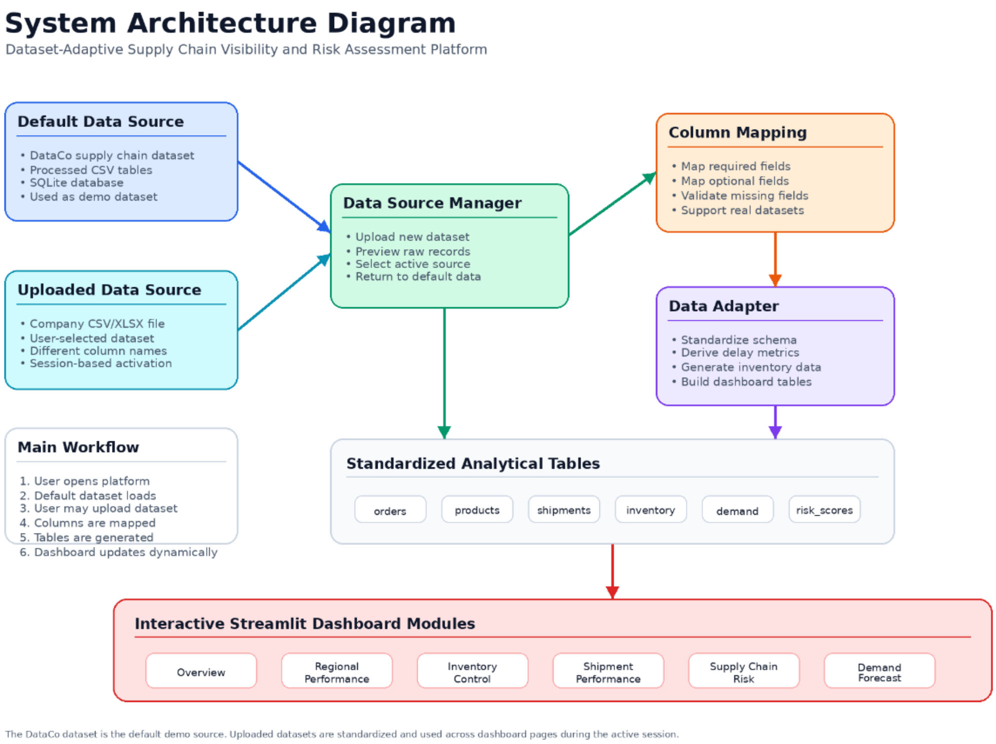

Figure 1. System architecture of the dataset-adaptive supply chain visibility and risk assessment platform.

As shown in Figure 1, the uploaded dataset does not directly replace the dashboard tables. First, it passes through the Data Source Manager and column mapping process. Then, the data adapter converts it into the standard analytical structure used by the dashboard. This design allows the same dashboard pages to work with different supply chain datasets.

7.2 Data Source Manager and Dataset Upload

The Data Source Manager is the main component that makes the platform reusable. When the dashboard is opened, the default DataCo dataset is active. If the user wants to analyze another supply chain dataset, the user can upload a CSV or Excel file from the Data Source Manager on the Overview page.

After uploading a dataset, the system shows a preview of the raw data. Then, the user maps the uploaded columns to the standard supply chain schema. This mapping process is necessary because different datasets may use different column names for the same concept. For example, one dataset may use "Order Date," while another dataset may use "PO Created At."

The uploaded dataset is used during the active Streamlit session. If the user activates the uploaded dataset, all dashboard pages use the uploaded data. If the user wants to return to the default dataset, the Data Source Manager provides an option to return to the default DataCo dataset.

7.3 Standard Data Schema

The standard data schema defines the common structure used by the platform. The uploaded dataset must be mapped to this schema before it can be used by the dashboard pages. Some fields are required, while others are optional. Required fields are necessary for basic dashboard generation. Optional fields improve the quality of risk analysis, shipment performance analysis, and financial indicators.

The standard schema used in the platform is shown in Table 2.

Table 2: Standard Supply Chain Schema Used in the Platform

Standard Field	Required	Description
order_id	Yes	Unique order or transaction identifier
order_date	Yes	Date when the order was created
product_name	Yes	Product, item, material, or SKU name
category_name	Yes	Product category or product group
order_quantity	Yes	Ordered quantity
shipping_date	No	Actual shipping, dispatch, or delivery date
scheduled_delivery_date	No	Planned or scheduled delivery date
delivery_status	No	Delivery status such as on time or late delivery
actual_shipping_days	No	Actual shipping or transit duration in days
scheduled_shipping_days	No	Planned or promised shipping duration in days
shipping_mode	No	Transport mode or shipment method
market	No	Market or business area
order_region	No	Region related to the order or delivery
order_country	No	Country related to the order or delivery
sales	No	Sales, revenue, or order value
order_profit	No	Profit, margin, or financial result

If an optional field is not available in the uploaded dataset, the system uses default values or derives indicators when possible. For example, if actual shipping days and scheduled shipping days are not available, the system can calculate delay days by comparing the actual delivery date and the scheduled delivery date. This improves the flexibility of the platform and allows it to work with different supply chain datasets.

7.4 Dashboard Data Model

After the dataset is standardized, the system creates analytical tables that are used by the dashboard pages. These tables are not designed as a full enterprise resource planning database. Instead, they are designed as dashboard-ready analytical tables for visibility, reporting, and risk assessment.

The main analytical tables are orders, products, shipments, inventory, demand, and risk_scores. The orders table is the central table because it contains the main order-level information. The shipments table is created from order and delivery-related fields. The inventory table is generated from product and demand patterns. The demand table summarizes monthly demand by product category. The risk_scores table stores regional risk indicators.

The logical data model used by the dashboard is shown in Figure 2.

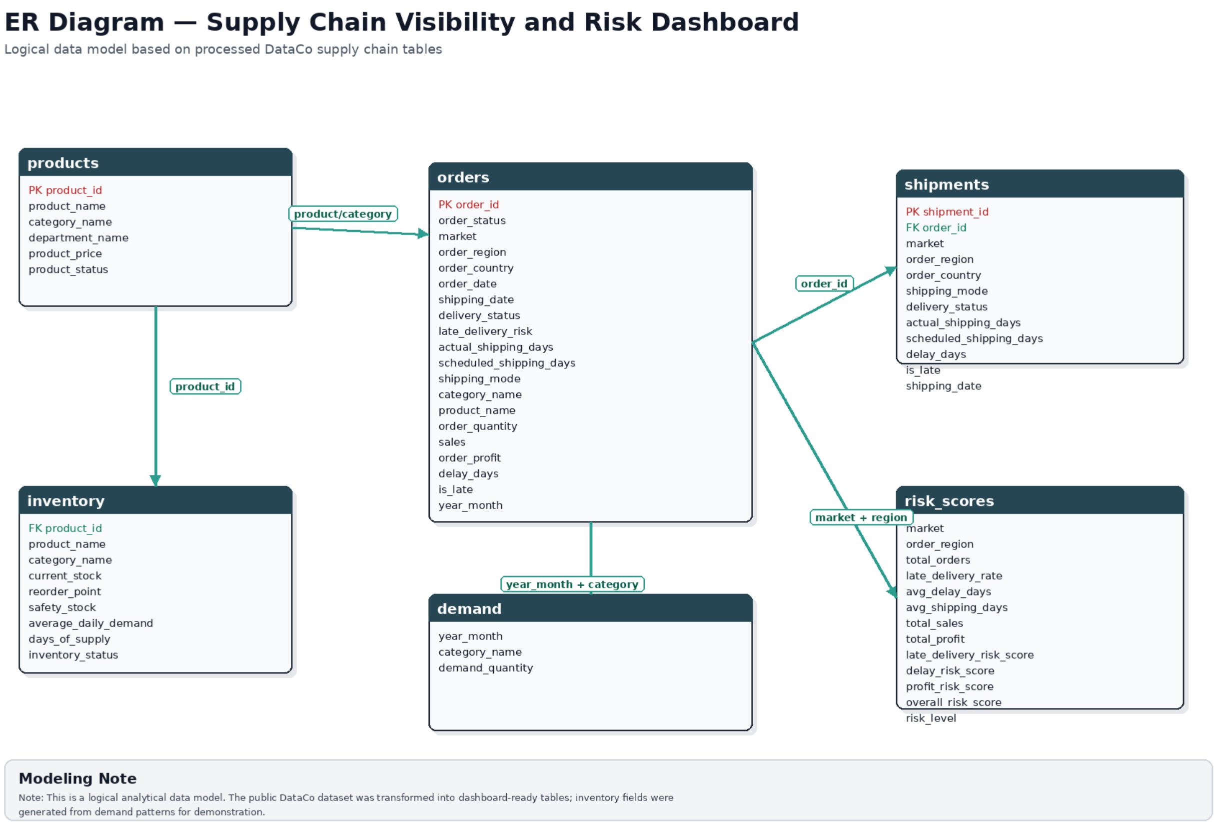

Figure 2. Logical data model used by the dashboard.

The logical data model represents the standardized analytical structure of the platform. Uploaded datasets are first converted into this structure by the data adapter. Therefore, even if the uploaded dataset has different column names, the dashboard can use the same analytical tables after the standardization process.

7.5 Dashboard Modules

The platform includes six dashboard modules. Each module focuses on a different part of supply chain visibility and risk analysis. These modules work with the active dataset. If the user activates an uploaded dataset, the same dashboard modules are updated according to the uploaded data.

The dashboard modules and their main outputs are shown in Table 3.

Table 3: Dashboard Modules and Their Main Outputs

Module	Main Purpose	Example Outputs
Overview	Provides general supply chain visibility	KPIs, monthly order volume, delivery status distribution
Regional Performance	Compares markets, regions, and countries	Regional KPIs, late delivery rate by market, regional risk
Inventory Control	Monitors product-level inventory indicators	Critical items, warning items, days of supply, inventory risk
Shipment Performance	Analyzes delivery and shipment performance	Late delivery rate, delay by shipping mode, shipment trend
Supply Chain Risk	Combines operational risk indicators	Overall risk score, delivery risk, inventory risk, demand volatility risk
Demand Forecast	Analyzes historical demand and future demand estimates	Moving average forecast, linear regression forecast, MAE, MAPE

The modular structure makes the platform easier to understand and maintain. Each page has a specific responsibility, but all pages use the same active dataset. This design supports both the default DataCo dataset and user-uploaded datasets.

7.6 Risk Scoring Logic

The supply chain risk module uses a rule-based and interpretable risk scoring approach. The aim is to provide clear risk indicators that can be understood by users. The model does not use a black-box machine learning algorithm for risk scoring. Instead, it combines operational indicators such as delivery performance, delay days, inventory status, and demand volatility.

The main risk components used in the platform are shown in Table 4.

Table 4: Risk Components Used in the Overall Risk Score

Risk Component	Meaning	Weight
Delivery Risk	Share of late deliveries in the selected dataset	35%
Inventory Risk	Weighted share of critical and warning inventory items	25%
Delay Risk	Average delay days normalized with 3 or more days treated as maximum delay risk	20%
Demand Volatility Risk	Variation in monthly demand	20%

The overall risk score is calculated by combining the weighted risk components. The weights were selected to give higher importance to delivery performance and inventory condition, because these two areas directly affect operational continuity and customer service.

The platform uses rule-based risk scores to provide an interpretable assessment of supply chain performance. The overall supply chain risk score is calculated from four main components: delivery risk, inventory risk, delay risk, and demand volatility risk. In addition to the overall score, regional and category-level risk tables use related operational indicators to compare risk across regions and product categories.

$$Late\ Delivery\ Rate = \left(\frac{Number\ of\ Late\ Orders}{Total\ Number\ of\ Orders}\right) x\ 100 \tag{1}$$

$$Average\ Delay\ Days = \frac{Total\ Delay\ Days}{Number\ of\ Orders} \tag{2}$$

$$Delay\ Risk = \min\left(\left(\frac{Average\ Delay\ Days}{3}\right) x\ 100, 100\right) \tag{3}$$

$$Inventory\ Risk = \left(\frac{(Critical\ Items\ x\ 1.0) + (Warning\ Items\ x\ 0.5)}{Total\ Inventory\ Items}\right) x\ 100 \tag{4}$$

$$Demand\ Volatility\ Risk = \left(\frac{Standard\ Deviation\ of\ Monthly\ Demand}{Average\ Monthly\ Demand}\right) x\ 100 \tag{5}$$

The overall supply chain risk score is then calculated as:

$$\begin{aligned} Overall\ Risk\ Score = {} & 0.35\ x\ Late\ Delivery\ Rate + 0.25\ x\ Inventory\ Risk \\ & + 0.20\ x\ Delay\ Risk + 0.20\ x\ Demand\ Volatility\ Risk \end{aligned} \tag{6}$$

In this prototype, an average delay of three days or more is treated as the maximum delay risk. Critical inventory items are treated as full-risk items, while warning inventory items are treated as half-risk items. This makes the inventory risk score more balanced than treating all warning items as severe as critical items.

The risk score is interpreted in three levels. A score below 30 is classified as Low Risk, a score between 30 and 60 is classified as Medium Risk, and a score above 60 is classified as High Risk. This classification helps users understand the risk level quickly without analyzing every component separately.

The displayed risk scores are also affected by dashboard filters. When the user filters the dataset by market, category, shipping mode, or date range, the risk components are recalculated based on the filtered data. This makes the risk analysis interactive and more useful for decision-making.

7.7 Demand Forecasting Logic

The Demand Forecast module uses historical monthly demand values generated from the active dataset. The demand table is created by grouping order quantity values by year-month and product category. This allows the system to analyze demand trends over time.

The platform uses two simple forecasting methods: 3-month moving average and linear regression. The moving average method is used to show short-term demand trends based on recent historical values. The linear regression model is used to estimate future demand by learning a simple trend from historical monthly demand. (scikit-learn developers, n.d.)

The forecasting module also includes performance evaluation metrics such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). These metrics help users understand how close the forecasted values are to actual historical demand values.

The forecasting approach is intentionally simple because the main focus of the project is supply chain visibility and risk assessment. The purpose of the forecast module is to demonstrate how demand data can support planning decisions, not to create a highly advanced forecasting system.

7.8 Dashboard Interface and Screenshots

The dashboard interface was developed with Streamlit. The platform includes a sidebar navigation menu and multiple dashboard pages. The Overview page includes the Data Source Manager, which allows users to upload a new dataset, map columns, activate the uploaded dataset, or return to the default DataCo dataset.

The Data Source Manager is shown in Figure 3.

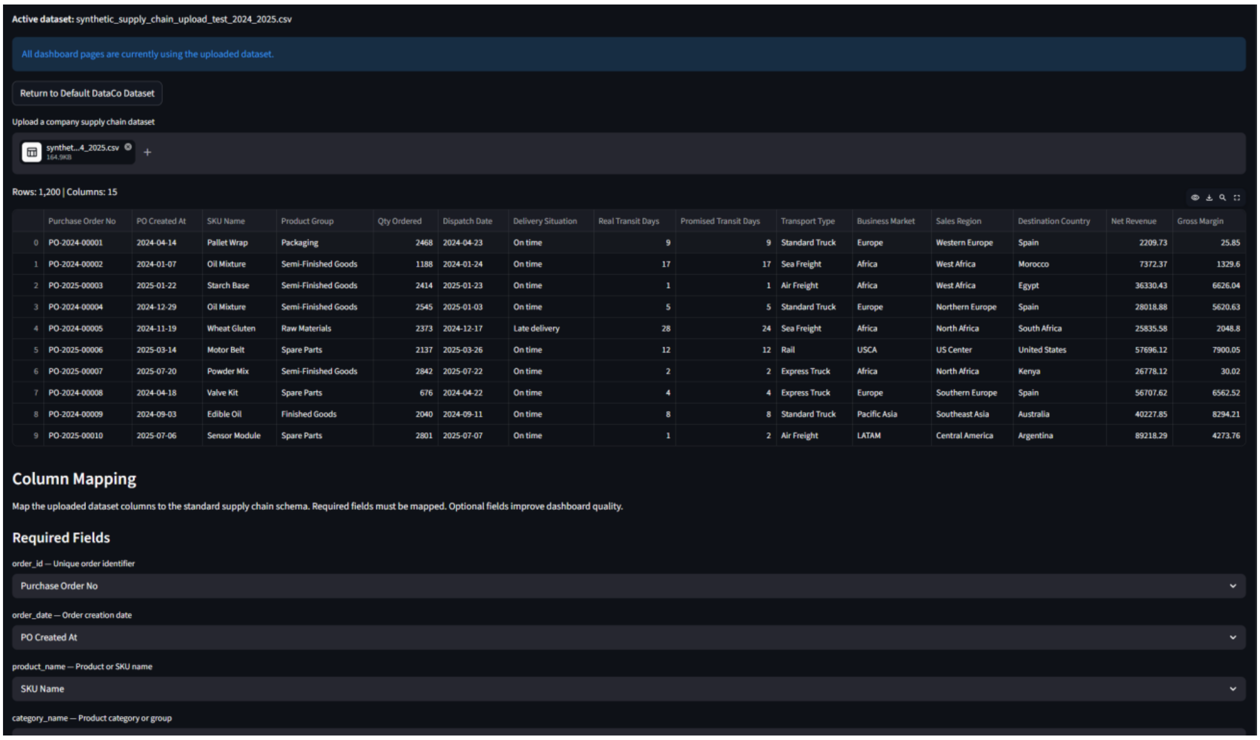

Figure 3. Data Source Manager and uploaded dataset workflow.

As shown in Figure 3, the user can upload a CSV or Excel dataset and map it to the standard supply chain schema. After the uploaded dataset is activated, all dashboard pages use the uploaded data during the active session.

The Overview page provides a high-level summary of the active dataset. It shows key performance indicators, monthly order volume, delivery status distribution, average delay by shipping mode, inventory status summary, and risk level distribution.

An example of the Overview dashboard generated from an uploaded dataset is shown in Figure 4.

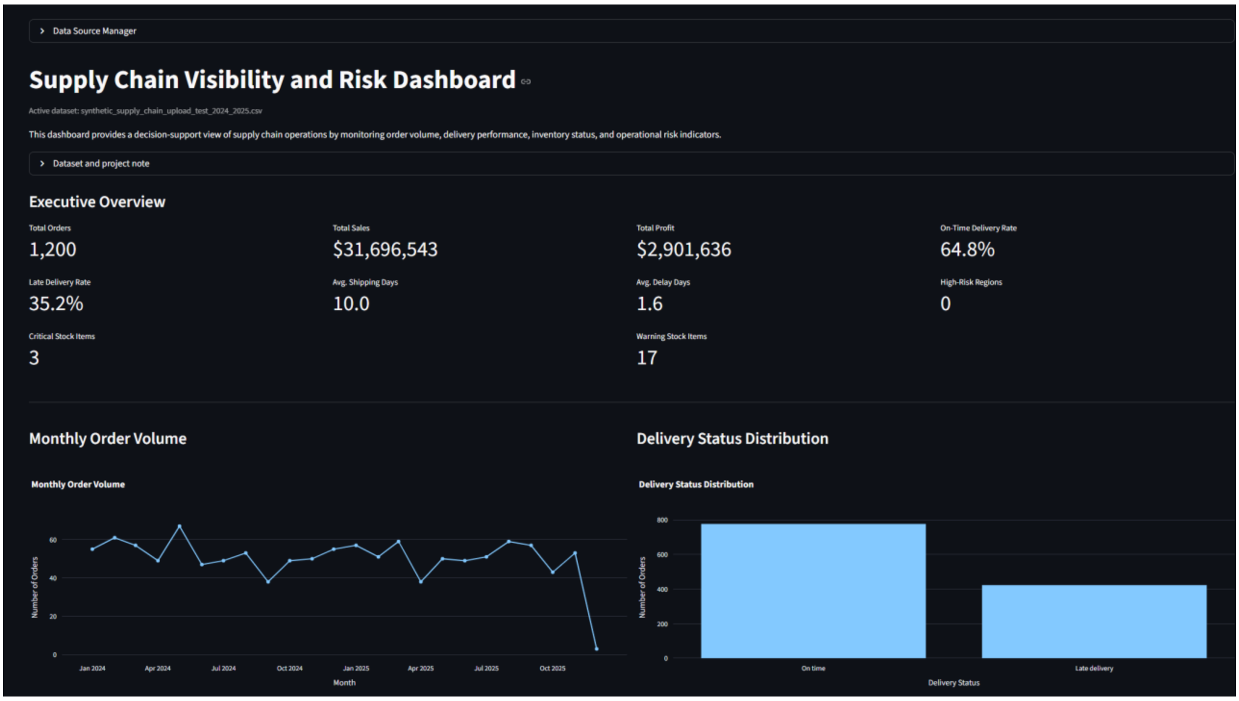

Figure 4. Overview dashboard generated from an uploaded dataset.

The Supply Chain Risk page combines several risk indicators into a single decision-support view. It includes delivery risk, delay risk, inventory risk, demand volatility risk, and an overall risk score. The risk results change according to the selected dashboard filters.

An example of the Supply Chain Risk dashboard is shown in Figure 5.

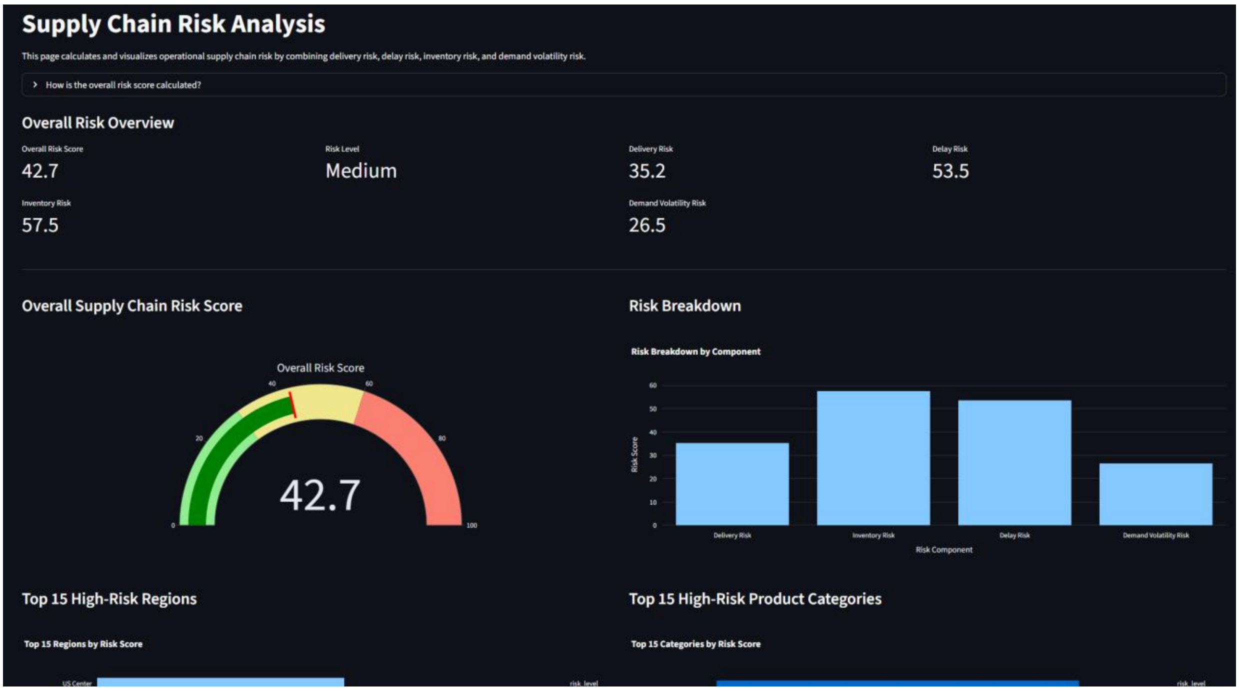

Figure 5. Supply Chain Risk dashboard generated from an uploaded dataset.

The Demand Forecast page analyzes historical demand and generates future demand estimates. It allows users to compare actual demand with forecasted values and review forecast error metrics.

An example of the Demand Forecast dashboard is shown in Figure 6.

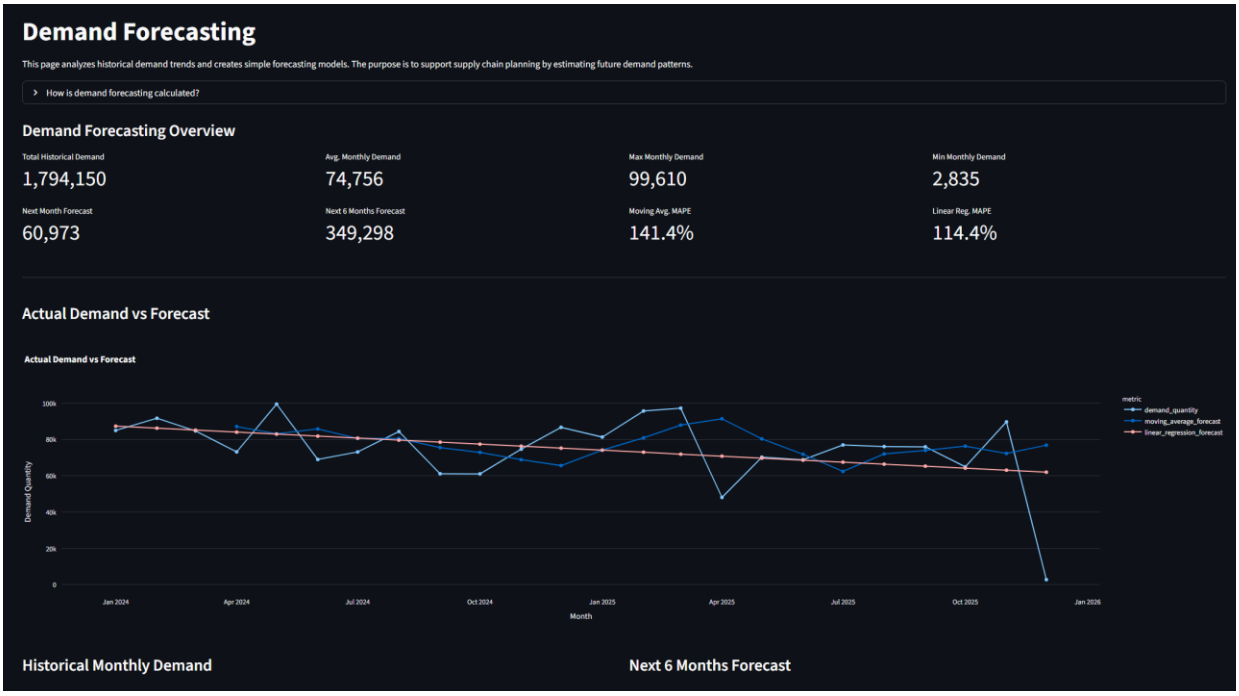

Figure 6. Demand Forecast dashboard generated from an uploaded dataset.

These dashboard pages show that the platform is not limited to a static dataset. When a new dataset is uploaded and activated, the same dashboard structure is regenerated based on the active dataset. This supports the main goal of the project: creating a reusable supply chain visibility and risk assessment platform.

8 CASE STUDY / BUSINESS CASE

This section presents the business case of the project. The platform is designed for a company that needs to monitor supply chain performance, identify operational risks, and analyze demand patterns from a single interface.

8.1 Business Scenario

The business scenario represents a company that manages products, orders, shipments, inventory indicators, markets, regions, and demand information. In this type of company, supply chain data may come from different systems or files. For example, order information may be stored in one file, shipment information may have different column names, and delivery performance may be reported separately.

The company needs a reusable dashboard platform that can work with different supply chain datasets. The platform should not require code changes every time a new dataset is analyzed.

Instead, users should be able to upload a dataset, map its columns to a standard schema, and generate dashboard pages based on the active dataset.

In this project, the DataCo dataset is used as the default demonstration dataset. However, the business case also considers a situation where a user uploads a new company dataset. After the uploaded dataset is mapped and standardized, the dashboard pages are updated dynamically.

8.2 User Roles

The platform can support different user roles in a supply chain environment. Each role may use the dashboard for a different purpose. The main user roles are shown in Table 5.

Table 5: User Roles and Dashboard Usage

User Role	Main Need	Related Dashboard Pages
Supply Chain Manager	Monitor overall performance and risk level	Overview, Supply Chain Risk
Logistics Analyst	Analyze shipment delays and delivery performance	Shipment Performance, Regional Performance
Inventory Planner	Identify critical and warning inventory items	Inventory Control, Supply Chain Risk
Business Analyst	Review trends, regional results, and demand patterns	Overview, Regional Performance, Demand Forecast

8.3 Use Case Structure

The main use cases of the platform are related to dataset upload, column mapping, dataset activation, dashboard monitoring, risk analysis, and forecasting. A user can either work with the default dataset or upload a new dataset. If a new dataset is uploaded, the user maps the dataset columns to the standard schema and activates the uploaded dataset. After activation, the dashboard pages use the uploaded dataset during the active session.

The use case diagram of the platform is shown in Figure 7.

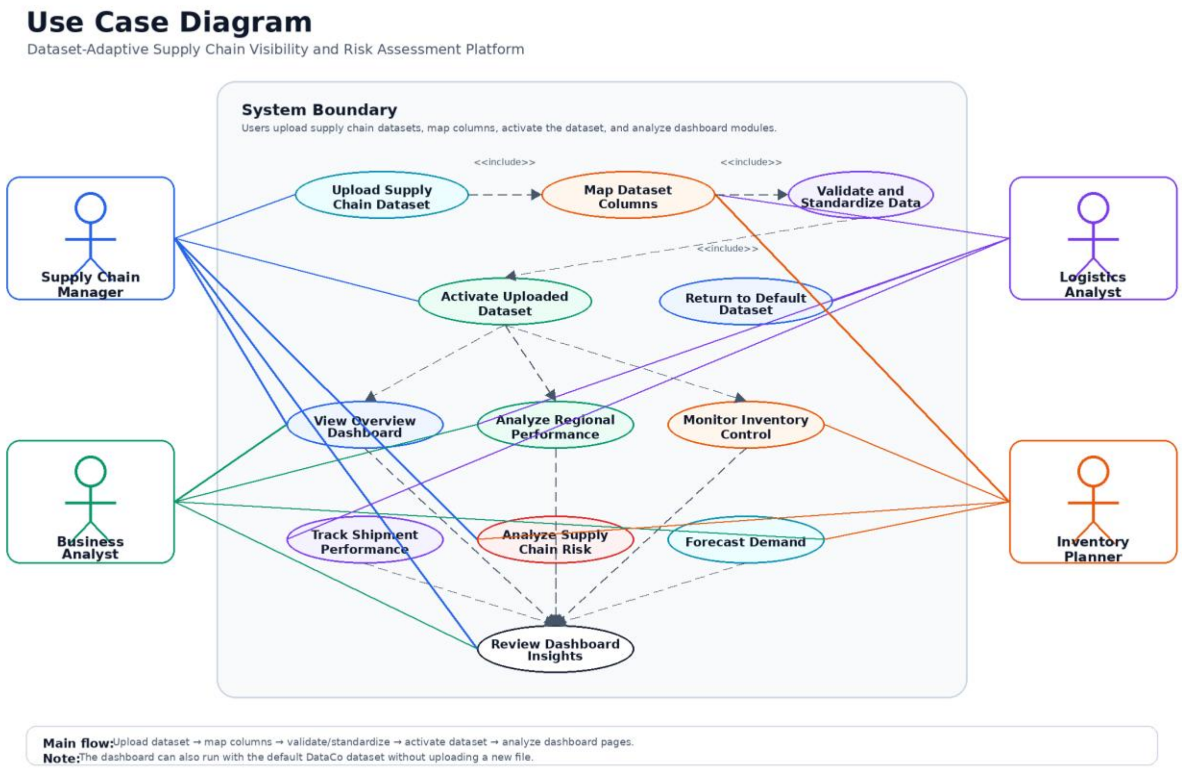

Figure 7. Use case diagram of the dataset-adaptive dashboard platform.

As shown in Figure 7, the dataset upload process is one of the central use cases of the platform. The user can upload a supply chain dataset, map its columns, validate and standardize the data, and activate it for dashboard analysis. After this process, the user can view the dashboard modules and review operational insights.

8.4 Example Usage Flow

An example usage flow starts with the user opening the platform. At first, the default DataCo dataset is active. The user can review the default dashboard, or upload a new supply chain dataset through the Data Source Manager.

If the user uploads a new dataset, the system displays a preview of the raw data. The user then maps the uploaded columns to the standard schema. Required fields such as order ID, order date, product name, category name, and order quantity must be mapped. Optional fields such as shipping mode, scheduled delivery date, sales, and profit improve the quality of the analysis.

After the user activates the uploaded dataset, all dashboard pages are updated according to the uploaded data. The user can then analyze regional performance, inventory status, shipment delays, supply chain risk, and demand forecasts. This process shows how the platform can support different supply chain datasets without changing the source code.

8.5 Business Value

The main business value of the platform is that it improves supply chain visibility and supports operational risk assessment. Instead of checking separate reports, users can analyze key supply chain indicators from one dashboard structure.

The dataset upload and standardization feature also increases the reusability of the system. A company can test different datasets by mapping them to the standard schema. This makes the platform more flexible than a dashboard designed for only one fixed dataset.

The platform can help users identify late delivery problems, risky regions, critical inventory items, delay patterns, and demand changes. These insights can support better planning and faster decision-making in supply chain operations.

8.6 Testing and Validation

The developed platform was tested by using both the default DataCo supply chain dataset and the uploaded dataset workflow. The purpose of testing was to verify that the data processing pipeline, dashboard modules, risk scoring logic, and demand forecasting outputs work correctly with the standardized data structure.

First, the default DataCo dataset was processed through the data preparation script. The raw dataset contained 180,519 order records, and the script generated dashboard-ready analytical tables such as orders, shipments, products, inventory, demand, and risk scores. These processed files were then used by the Streamlit dashboard pages.

Second, the uploaded dataset workflow was tested through the column mapping interface. Required fields were mapped to the standard schema, while optional fields were either mapped when available or handled through default and derived values. After the uploaded dataset was activated, the dashboard modules were generated based on the active standardized dataset.

The main validation checks are summarized below.

Table 6: Summary of Functional Testing and Validation Results

Test Case	Expected Result	Actual Result	Status
Default dataset processing	Processed CSV files and SQLite database are generated	Generated successfully	Passed
Dashboard module loading	Overview, regional performance, inventory, shipment, risk, and demand pages load without syntax errors	Dashboard modules are available	Passed
Uploaded dataset mapping	Required columns are mapped to the standard schema	Dataset can be standardized and activated	Passed
Missing optional fields	Missing optional fields are handled with default or derived values	Dashboard remains usable	Passed
Risk scoring logic	Overall risk score uses late delivery rate, inventory risk, delay risk, and demand volatility risk	Rule-based risk score is calculated and displayed	Passed
Demand forecasting	Moving average and linear regression forecasts are generated from monthly demand data	Forecast tables and charts are displayed	Passed

These tests show that the platform can process the default dataset, support uploaded datasets through schema mapping, and generate the main dashboard outputs. However, the validation is limited to functional testing. The current version does not include automated unit tests, stress testing with very large uploaded files, or live ERP/API integration testing. These limitations are considered future improvement areas.

9 CONCLUSIONS

This graduation thesis developed a Supply Chain Visibility and Risk Assessment Platform using Python and Streamlit. The main goal of the project was to create an interactive decision-support system that can help users monitor supply chain performance, identify operational risks, and analyze demand patterns from a single interface.

The completed platform includes several dashboard modules: Overview, Regional Performance, Inventory Control, Shipment Performance, Supply Chain Risk, and Demand Forecast. These modules provide key performance indicators, charts, tables, risk scores, and forecasting outputs. The dashboard helps users analyze delivery performance, shipment delays, inventory status, regional differences, demand trends, and overall supply chain risk.

One of the strongest parts of the project is the dataset upload and standardization feature. The system is not limited to the default DataCo dataset. Users can upload a new supply chain dataset in CSV or Excel format, map the dataset columns to a standard schema, and generate dashboard pages based on the uploaded dataset. This makes the platform more reusable and flexible than a static dashboard built for only one dataset.

The project also demonstrates how supply chain data can be transformed into dashboard-ready analytical tables. The system creates or uses order, product, shipment, inventory, demand, and risk score tables. These tables support the dashboard modules and make it possible to analyze different parts of the supply chain in an integrated way.

The risk assessment approach is rule-based and interpretable. The platform calculates risk by using indicators such as late delivery rate, average delay days, inventory status, and demand volatility. This approach allows users to understand the main reasons behind the risk score. The forecasting module also supports planning by showing historical demand patterns and simple future demand estimates.

The project has some limitations. The default DataCo dataset is a public historical dataset and does not represent the current operations of a specific company. In addition, the quality of the dashboard outputs depends on the structure and completeness of the uploaded dataset. Some indicators, especially inventory-related fields, are generated for academic demonstration purposes when the uploaded dataset does not contain complete inventory information. The risk score is also rule-based rather than machine-learning-based.

In future work, the platform can be improved by adding automatic column detection, database persistence for uploaded datasets, exportable PDF risk reports, real-time ERP or API integration, and more advanced forecasting models. The risk scoring system can also be improved by using machine learning methods if enough historical labeled risk data is available.

Overall, the project successfully demonstrates a reusable and dataset-adaptive dashboard platform for supply chain visibility and risk assessment. It shows how different supply chain datasets can be standardized and transformed into operational insights through dashboards, risk indicators, and forecasting modules.

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Project GitHub Repository:
https://github.com/dolunayunlu/supply-chain-visibility-risk-dashboard

Default Dataset Source:
https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

The GitHub repository includes the source code, dashboard pages, data processing script, project diagrams, screenshots, README file, requirements file, and one-click Windows launcher file.

The default dataset is the DataCo Smart Supply Chain dataset. The platform also supports user-uploaded CSV and Excel supply chain datasets through the Data Source Manager.

12 CURRICULUM VITAE

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Education:

Bachelor's Degree: Piri Reis University

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Work Experience / Internship:

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Worked in an international environment and improved communication, teamwork, responsibility, and customer-facing problem-solving skills.

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2026

Completed applied coursework in Excel and Python for data analysis and reporting.

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Python, pandas, NumPy, Matplotlib, Excel, Power Query, SQL, Java, HTML, CSS, JavaScript, Cisco Packet Tracer, data cleaning, exploratory data analysis, reporting, dashboard development

Academic Projects:

- **House Prices – Advanced Regression Techniques, Kaggle Competition**

Performed data cleaning and exploratory data analysis on housing data using Python.

- **Animal Lifespan Prediction and Clustering**

Conducted data preprocessing and exploratory analysis on a biological dataset using Python.

- **Secure Office Network Design**

Designed a segmented office network in Cisco Packet Tracer using VLANs, DHCP, NAT, ACLs, and SSH.

- **Statistical Analysis of Student Performance**

Applied descriptive statistics, correlation analysis, hypothesis testing, and multiple linear regression.

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Developed a Java-based library management application using object-oriented programming principles.

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